

Заметки о поиске работы в промышленности

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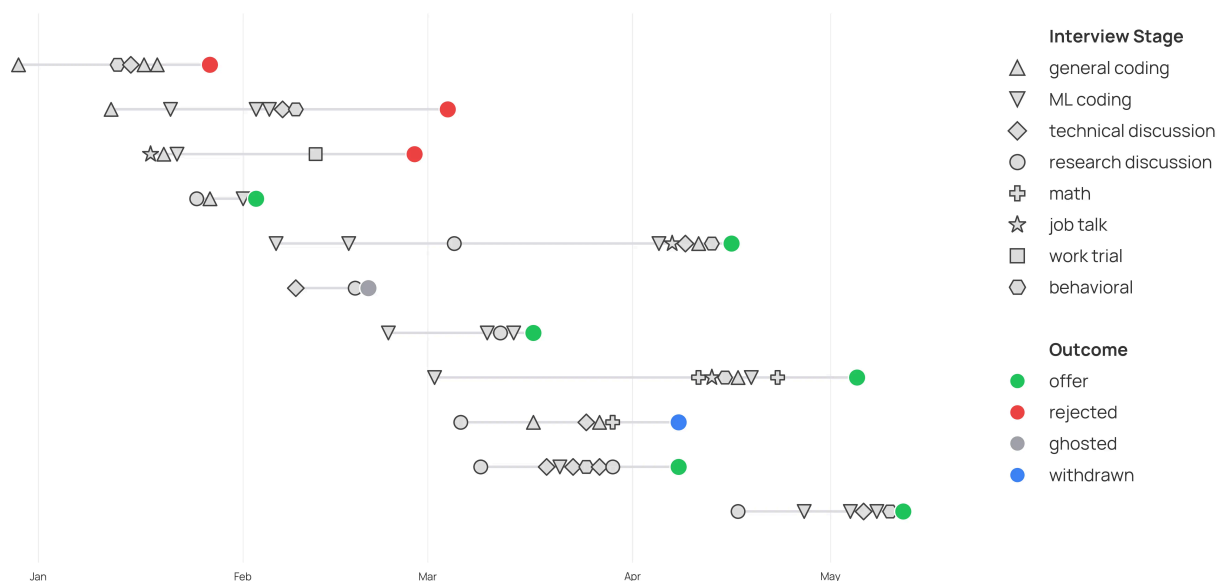
Большую часть времени, пока я работал над докторской диссертацией, поиск работы в моем сознании ассоциировался с сортировочной машиной 🚂: старшие аспиранты исчезали (на несколько месяцев), а потом появлялись, и их судьба была решена. Даже когда мои близкие друзья заканчивали учебу и устраивались на работу, я почти ничего не знал о том, что с ними происходило, если не считать редких вестей. Когда наконец подошла моя очередь, я обнаружила, что процесс оказался гораздо более сложным, чем я себе представляла, и мне казалось, что я изучаю правила игры прямо в процессе.

Оглядываясь назад, я понимаю, что многие из моих трудностей были типичными для всех, а многое из того, что я узнала на этом пути, теперь кажется общеизвестным. Я пишу этот пост, чтобы поделиться одним из примеров того, как может проходить этот процесс, и, надеюсь, сделать поиск работы чуть менее загадочным для тех, кто недавно был на моем месте.

Немного обо мне. В конце шестилетней работы над докторской диссертацией по обработке естественного языка в Вашингтонском университете я подал заявку на должность научного сотрудника / члена технического персонала. Я учился в университете всю свою жизнь и с удовольствием остался бы в аспирантуре, но мои научные руководители в конце концов убедили меня двигаться дальше. Большую часть времени, пока я работал над диссертацией, я не особо задумывался о том, что буду делать после ее защиты, и меня больше привлекали интересные идеи, чем что-либо еще. Это привело к частым сменам направления деятельности, но, к счастью, за последние два года мне удалось выработать последовательный подход (к токенизации!) потому что это совпадало с моим увлечением, и я думаю, что наличие узкой специализации помогло мне выделиться при поиске работы.

Моя хронология

На рисунке ниже показана моя хронология поиска работы (по мотивам [поста Натана Ламберта](#)). Серой заливкой обозначены собеседования, а цветными кружками — результаты. Обратите внимание, что **закрашенные** означает, что рекрутер так и не сообщил мне о результатах или дальнейших шагах, а **отмеченные** — что я вежливо сообщил компании, что больше не заинтересован в работе, после того как получил несколько предложений, которые меня заинтересовали. В общей сложности я прошел собеседования в 11 компаниях, всего было 57 собеседований. Не показаны 46 дополнительных звонков от рекрутеров и 16 чатов после получения предложения о работе, а также множество неформальных бесед с потенциальными работодателями, предшествовавших поиску работы.



По заказу компании. Я решал, когда начинать процесс собеседования, исходя из нескольких факторов: чувствовал ли я себя готовым, оказывала ли на меня давление компания, как быстро, по моему мнению, они должны были принять решение, насколько я был заинтересован в работе, а также из менее осознанных факторов, таких как прокрастинация. Обычно для практики выбирают несколько компаний, а затем распределяют остальные процессы так, чтобы все предложения поступали примерно в одно и то же время для удобства переговоров. Я считаю, что в целом это правильный подход, но хотел бы добавить несколько замечаний.

- Практикуйтесь в прохождении собеседований, но помните, что ваша выносливость не безгранична — не переусердствуйте, чтобы не перегореть к

тому моменту, когда вы доберетесь до тех мест, которые вам действительно интересны!

- Есть внешние факторы, которые стоит учитывать при определении сроков, например наличие у компании штатного расписания и то, какие команды активно набирают сотрудников. Это может оказаться важнее вашей подготовки. Вы можете узнать об этом от своих друзей и рекрутеров.
- Сроки не имеют большого значения, поэтому не стоит привязываться к конкретным датам. Рекрутеры понимают, что у вас есть и другие дела, и используют различные уловки, чтобы отсрочить получение предложения о работе и принятие решения. Тем не менее бывают и вопиющие исключения (так называемые «взрывоопасные» предложения), поэтому важно выяснить, сколько времени обычно дают кандидатам на подписание документов.

Getting the first interview. To state the obvious: try to do good work during the PhD, make friends, and collaborate a lot! To get that first interview, sometimes you need to have someone inside the company vouching for you. You can set yourself up for success early on by being social at conferences, collaborating widely, and attending networking events (of course this part doesn't come easily to everyone — certainly not for me — so take care of your own energy and comfort levels too). During the job search, reach out to people you know (or don't know) and ask about opportunities. In fact, a big part of the job search is reconnecting with people who you may not have talked to in years — this is okay, expected, and turns out to be a wonderful side effect of the process.

Interview types

I would say there were roughly the following categories of interviews. Overall, technical skills and knowledge are evaluated much more than research experience, though the latter probably gets you the interview in the first place.

ML coding. This was by far the most common. These questions may ask you to implement a given architecture, a decoding strategy, a traditional ML algorithm, or sometimes way more creative things. Being fluent in PyTorch is a must; in rare cases I was asked to use only numpy, for instance when writing the backwards pass from scratch, but I was not expected to be familiar with the numpy syntax.

General coding. Basically LeetCode, sometimes with some extra flavor. It's good to build strong foundations here because the concepts often show up in ML coding interviews, too.

Technical discussion. These interviews do not involve coding but are very much technical. Sometimes, the interview is an extended discussion around one topic, such as how you would design experiments to answer a particular research question or accomplish a particular goal. The interviewer will generally press you on your design choices and ask you to comment on some hypothetical results and design follow-up experiments. In other cases, the interview consisted of a list of rapid-fire questions (*What are some different ways of encoding positional information? What is 5D parallelism? What is the difference between PPO and GRPO?*), and the goal was to signal that I knew my stuff. The former type of interview tests how you think, whereas the latter checks your breadth of knowledge on the field.

Research discussion. These are the kinds of conversations we practiced most in our PhD. The interviewer generally asks you to start by telling them about a past project, and the rest of the discussion flows from there. They might also ask questions about other papers on your CV. When preparing for these kinds of discussions, it's useful to take a step back and think about why you chose to work on the things you did, insights and opinions you've developed along the way, and what you view as promising future directions. I also tailored my research pitch depending on the role; interviewers are tired, so hitting the right keywords makes it easier for them to believe that your profile is relevant.

Behavioral. These are totally textbook behavioral interviews, apart from the occasional question about AI safety or societal impacts. Enumerate memorable stories from your PhD and map them onto the common behavioral questions so that during the interview, you can retrieve the right anecdotes instantly. I failed my first behavioral interview because I went into it thinking I'm obviously well-"behaved," and came up blank on excruciatingly simple questions. Trust me, it is uniquely painful to try to reconstruct hazy memories at the same time as delivering them in an interview, only for the interviewer to say at the end, "You didn't answer the question."

Math. Some companies have a math interview, ranging from fun logic puzzles to serious mathematical derivations with pen and paper. I would recommend brushing up on probability, linear algebra, and calculus.

Job talk. There is some variation in what the job talk looks like, but compared to an academic one, it tends to be a bit shorter and focused on a single paper or direction. My job talk was all about tokenizers; I spent most of the time on a first-author work and then covered a few second-author and ongoing works briefly, as fortunately they tied together very nicely.

Preparation

There is truly no better use of your time than studying for interviews. For me, the experience was very much like being back in undergrad: I took notes (see my [LLM notes](#), which I worked on continuously throughout the process, and my [math notes](#), which was all for a single fateful interview), drew diagrams, did practice problems, and spent entire days in coffee shops making sure I understood fundamental ML concepts inside-and-out. Technical interviews are hard, and the skills being tested require dedicated effort to develop outside of doing research. For me and for most people I talked to, the job search is a full-time job.

I started my process by watching all the lectures from Stanford's [Language Modeling from Scratch](#) course, which is helpful for illustrating the breadth of topics I needed to learn and helped me organize many scattered concepts in my brain into one coherent picture of the field. After covering the basics, I spent the rest of my time deep-diving concepts one at a time by reading relevant blog posts & papers, talking to ChatGPT & Claude *a lot*, and practicing implementing things from scratch. [Homework 1](#) is crucial: implementing / debugging a transformer comes up so often in interviews that it will pay off massively to turn it into muscle memory and really isn't worth losing points on. Make sure you are practicing coding with AI assistance completely off to mimic interview settings (you will underestimate your reliance otherwise)!

I found that each interview is unique and can benefit from a little — sometimes a lot — of dedicated preparation. You can usually build an intuitive understanding of an interview's scope from the provided description, the topics that the company is interested in, hints from the recruiter, and the reputation of the company. When I was in the thick of interviewing, I found that I was constantly swapping information in and out of my brain so that the most relevant knowledge for a particular interview would be fresh. The best way I can describe it is: each interview is a slightly different math or CS class, you never went to lectures, and now you have ~3 days to cram for the midterm.

Day of interview. Perhaps it is because I am getting old, but nothing beats getting enough sleep the night before the interview. I made the mistake of doing my first technical interview on 2 hours of sleep after cramming all the intricacies of LLM inference into my brain — none of the last-minute knowledge came up, and I ended up spending 10 minutes on an off-by-one error because my gears were barely turning. After the interview, remember to record some notes, which will be helpful for your future studying and reflection.

Side benefits. Studying carried enormous side benefits for me. Having a wider breadth of knowledge directly improved my confidence as a researcher. I became more secure in conversations because I was less worried about gaps in my knowledge being exposed, and no longer felt compelled to hide them when they came up. I truly believe that if I had done some of this studying earlier in my PhD, it would have expanded the space of problems I might be able to think about and have ideas in, and certainly the number of conversations I would have sought out. Amazingly, I also found that studying made me enormously more effective at my ongoing project. I was able to have technical ideas that I never would have been able to access before and do more technical work, which was thrilling.

Negotiation

I was shocked to learn that the work is not nearly done after you receive your offers. Instead, there is a (potentially extended) period of time for you to learn more about your options and negotiate your offers. It involves many conversations with potential future teammates / managers, lunch visits, and recruiter calls. At this stage I was managing an overwhelming amount of communication, and there were always emails I was guilty of not responding to.

The truth is that negotiating is hard. Nothing in our PhD prepared us for this, and unlike interviews, this part cannot be conquered by studying. Compared to recruiters, you are outmatched in both knowledge of the market and the skill of negotiation, and everyone you talk to wants something different from you. You may be thinking, “I would be happy with my offer and make a decision independently of compensation!”, and indeed it’s great to know your own values! But you’d be doing yourself a disservice if you didn’t negotiate. Initial offers leave room for negotiation by design; recruiters often explicitly invited me to play the game by saying things like, “I don’t expect you to take our first offer.” Putting in energy here for a few weeks can, literally, be equivalent to years of work at the initial offer.

It is really crucial at this stage to lean on your friends for the know-how of interacting with recruiters and for more data points to help calibrate your asks. Before every recruiter call, I wrote down what I was willing and not willing to share, along with quotes I could recite verbatim. In the post-offer stage, I would anticipate questions they might ask and points they might make, and carefully construct responses that I could deliver comfortably while still advocating for myself. Though time-consuming, it is really worthwhile to be deliberate about every aspect of the process.

Concluding words

In this blog post I focused on the concrete parts of the job search, but in reality a huge part of my personal experience was managing all the emotions that come with being on the market. There is a lot of social perception to navigate: it is not a good feeling to compare yourself to your peers, everyone has opinions on where you should or shouldn't go, and people become unusually invested in how your life is going. I also found it stressful navigating a huge decision space with incomplete information, where small choices with no right or wrong answers (like who to contact when) have an outsized impact. Frankly, I was stressed, miserable, and not functioning in other parts of my life for several months. Hopefully you find more joy, but if not, just know that you are not alone.

I've been hurtling towards the end of my PhD for months, and now at the end of it all, I'm immensely sad to leave this chapter of my life behind. The PhD is such a special time, where our only job is to have good ideas and execute them, to learn and grow as researchers, *without* worrying about imminently securing a real job. So while I hope this post helps you mentally prepare for the future (and I certainly recognize how distracting industry forces are today), I also hope that you can cherish your PhD for the unique time that it is. These goals may be complementary, after all — I consistently found that I did my best work when I was having fun and chasing the questions my mind would not lay to rest.

Appendix: learning resources

- [LeetCode 75 / NeetCode Blind 75](#)
- [Stanford CS336: Language Modeling from Scratch](#)
- [Self-Attention & Transformers](#)
- [The Illustrated GPT-2](#)

- [Backpropagation](#)
- [Introduction to Policy Gradient for LMs](#)
- [Lightweight Guide to understanding GRPO and RL principles](#)
- [How to Scale Your Model](#)